

Swapping LLM Providers

LESSON 03 · MODULE 1 — FOUNDATIONS

What you'll learn

- How a single string change moves an agent between providers
- Short-form vs explicit `provider/model` notation, and when to reach for each
- Which providers Lyzr supports out of the box
- The behavioural differences across OpenAI, Anthropic, Google, and Grok on the same prompt
- A practical default for picking a model when you start a new project

Key concepts

One string changes the brain. The `provider` parameter on `create_agent` takes a model name. Lyzr routes the call to the right place. Same agent, different brain.

Two notations. Short form is just the model name — `gpt-4o`, `claude-sonnet`, `gemini-2.5-pro`. Lyzr auto-detects which provider it belongs to. Explicit form is `provider/model` — `openai/gpt-4o`, `anthropic/claude-sonnet`. Start short; fall back to explicit only when an auto-detect error tells you to.

Providers supported out of the box: OpenAI, Anthropic, Google, Grok, Perplexity, AWS Bedrock.

Models converge on canonical answers. Four out of five providers in the side-by-side landed on "abstraction" as the most elegant idea in programming. That's the real lesson — for definition-shaped questions, the model choice barely matters. Provider comparisons earn their keep on **open-ended work** — creative angles, taste-dependent tasks. Not on questions with a single textbook answer.

Practical defaults. For most apps, start with the latest GPT or Claude Sonnet — broad capability, reliable, what most apps ship with. For high-volume work where each call doesn't need to be brilliant, drop to the latest GPT-mini, Claude Haiku, or Gemini Flash. For research and citations, Perplexity models are tuned for grounded answers with sources — different job. For most prototypes, the model choice matters less than the prompt; get the agent working on the default first, then try swapping providers.

Code shown in this lesson

PYTHON

TAKEAWAY

One string changes the brain.

TAKEAWAY

Two notations.

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Providers supported out of the box:.

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TAKEAWAY

Practical defaults.

```
# Short form – Lyzr auto-detects the provider
agent = studio.create_agent(provider="gpt-4o", ...)
agent = studio.create_agent(provider="claude-sonnet-4.5", ...)
agent = studio.create_agent(provider="gemini-2.5-pro", ...)

# Explicit form – provider/model
agent = studio.create_agent(provider="openai/gpt-4o", ...)
agent = studio.create_agent(provider="anthropic/claude-sonnet", ...)
```

PYTHON

```
# Same role, goal, instructions across four providers
prompt = "What's the most elegant idea in programming? Answer in two sentences."

for provider in ["gpt-4o", "claude-sonnet-4.5", "gemini-2.5-pro", "grok-llama-3.3-70b"]:
    agent = studio.create_agent(
        name=f"compare-{provider}",
        provider=provider,
        role="programming philosopher",
        goal="answer concisely",
        instructions="Two sentences max.",
    )
    print(provider, "→", agent.run(prompt).response)
```

TRY THIS

- 1 Run the same prompt — *"What's the most elegant idea in programming? Answer in two sentences."* — across five providers (GPT-4o, Claude Sonnet, Gemini Pro, Grok-Llama, and one Perplexity model). Document which providers converged on the same idea and which one disagreed.
- 2 Pick one provider you'd default to for *factual* queries and one for *creative* queries. Write two sentences explaining your reasoning. Keep it for reference the next time you start a project.